

A Divide-and-conquer Solution to 3D Human Motion Estimation from Raw MoCap Data

Jilin Tang*
NetEase Fuxi AI Lab

Lincheng Li†
NetEase Fuxi AI Lab

Jie Hou‡
NetEase Fuxi AI Lab

Haoran Xin§
NetEase Fuxi AI Lab

Xin Yu||
Faculty of Engineering and Information Technology

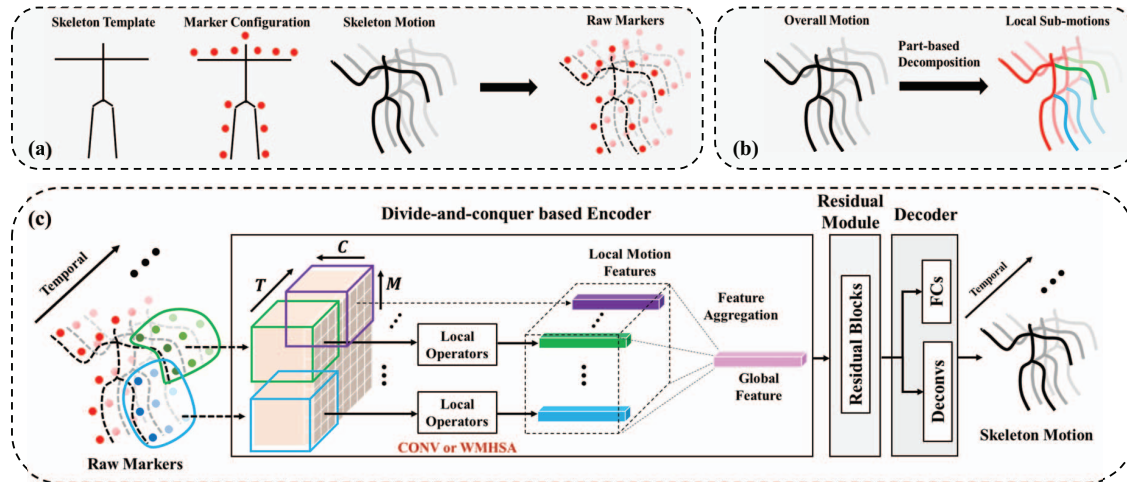


Figure 1: Overview of our proposed method. Top-left (a): the illustration of the motion capture procedure, where red points denote already labeled markers. Top-right (b): part-based motion decomposition, where different colors correspond to different local sub-motions. Bottom (c): the network architecture of our framework.

ABSTRACT

Marker-based optical motion capture (MoCap) aims to estimate 3D human motions from a sequence of input raw markers. In this paper, we propose a divide-and-conquer strategy based MoCap solving network that accurately retrieves 3D human skeleton motions from raw marker sequences in real-time. Our core idea is to decompose the task of direct estimation of global human motion from all markers into first solving sub-motions of local parts and then aggregating sub-motions into a global one to achieve accurate motion estimation.

Index Terms: Computing methodologies—Computer graphics—Animation—Motion capture; Computing methodologies—Computer graphics—Animation—Motion processing

1 INTRODUCTION

Marker-based optical motion capture (MoCap) enables users to control their avatars in virtual environments via their own 3D motions, which can be potentially applied in the field of virtual reality (VR). Technically, it utilizes multi-view optical sensors to track and record the 3D position coordinates of markers attached to the user's body.

However, the captured raw marker data inevitably contains some positional errors due to body occlusion, sensor precision limitation, etc. In a conventional workflow, dedicated technicians need to fix these marker errors manually, and then solve human skeleton motions from cleaned markers using a commercial software. However, the above process, especially the cleaning step, is extremely tedious and time-consuming [1, 2]. Thus, there is a high demand for automated algorithms to replace costly manual operations and achieve accurate and real-time MoCap cleaning and solving.

Recently, with the rapid development of deep learning, some researchers attempt to use data-driven neural networks to tackle the above problem. For example, Holden [2] proposed a simple fully-connected (FC) network with residual connections to directly predict the global transformations of human joints from the raw markers frame-by-frame. However, this method doesn't fully leverage the temporal information of MoCap data, which tends to break temporal continuity of motion sequences. To handle the above problem, Chen et al. [1] presented a two-stage framework, named MoCap-Solver, to predict skeleton motion sequences from sequential raw markers. Since they only consider the global correlation among all markers and do not effectively capture the spatio-temporal information of markers within local scopes, their method may generate sub-optimal results when complex motions occur.

In this work, we propose a divide-and-conquer strategy based MoCap solving framework to accurately estimate 3D human motions from sequential raw markers in real-time. As shown in Fig. 1(b), our key insight lies in the fact that the overall motion of a human body can be decomposed into several sub-motions of different local parts. Thus, instead of directly learning the overall motion from all markers as prior arts, our divide-and-conquer strategy firstly learns different

*e-mail: tangjilin@corp.netease.com

†e-mail: lilincheng@corp.netease.com

‡e-mail: houjie1@corp.netease.com

§e-mail: xinhaoran@corp.netease.com

||e-mail: xin.yu@uts.edu.au

local sub-motions from the corresponding marker subsets within the local spatio-temporal scopes, and then aggregates them into a global motion representation to achieve accurate motion estimation.

2 METHODOLOGY

Fig. 1(c) shows the whole pipeline of our network. It consists of a divide-and-conquer based encoder equipped with two different local operators, a residual module, and a decoder.

2.1 Divide-and-conquer Based Encoder

The encoder is the core component of our network, which employs a divide-and-conquer strategy to obtain the global motion representation effectively.

Divide-and-conquer Strategy. As shown in Fig. 1(c), we first partition all markers within the entire spatio-temporal dimensions into the overlapping local windows, where each window corresponds to one subset of all markers. Notably, the marker order in spatial dimension determines the division of marker subsets, i.e., different marker orders are equivalent to decomposing the overall motion of the human body into different combinations of local sub-motions, which may produce results with varying precision. Then, the encoder employs a series of local operators to capture local spatio-temporal relationships among different subsets of markers, i.e., each local operator aims to learn one certain local motion pattern. Furthermore, by performing the downsampling in the spatio-temporal dimensions, the encoder gradually aggregates local motion patterns into the global motion representation which contains the spatio-temporal information of the whole human motion sequence.

Local Feature Operators. Our encoder is agnostic against different local feature operators such as 2D local convolution or window-based multi-head self-attention (WMHSA) [3]. According to the types of local feature operators, our encoder can be implemented in two different versions: CONV-based and WMHSA-based. The CONV-based operator remains fixed after training while the WMHSA-based operator generates the weight parameters adaptively conditioned on different inputs. Specifically, the CONV-based encoder mainly contains one convolutional layer with stride = 1 and three convolutional layers with stride = 2, where the first layer uses 7×7 kernel and the other layers use 3×3 kernels. All convolutional layers are followed by batch normalization (BN) and ReLU activation. The WMHSA-based encoder roughly follows the architecture of Swin Transformer [3]. It mainly consists of one patch partition layer and three basic modules, where each module contains two successive Transformer blocks, followed by one patch merging layer which $2 \times$ downsamples the spatio-temporal resolution while $2 \times$ increasing the feature channels.

2.2 Subsequent Motion Predictor

The residual module and the decoder constitute our subsequent motion predictor, which aims to generate the complete human skeleton motion from the encoded global motion representation finally.

Residual Module. The residual module is introduced to equip our network with larger model capacity and stronger representation power. It consists of six residual blocks, where each contains two 3×3 convolutional layers with the same number of filters.

Decoder. The decoder aims to decode the intermediate motion features into the joint rotations and the root translations, respectively. As shown in Fig. 1(c), it contains two different branches. Specifically, the rotation branch consists of three deconvolutional layers with stride = 2 and one convolutional layer with stride = 1, where the first three layers use 3×3 kernels and the last layer uses 1×1 kernel. All layers are followed by BN and ReLU activation except the last output. The translation branch contains three FC layers, where the first two layers output 1024 hidden units followed by BN and ReLU activation, and the last layer directly outputs the root translations.

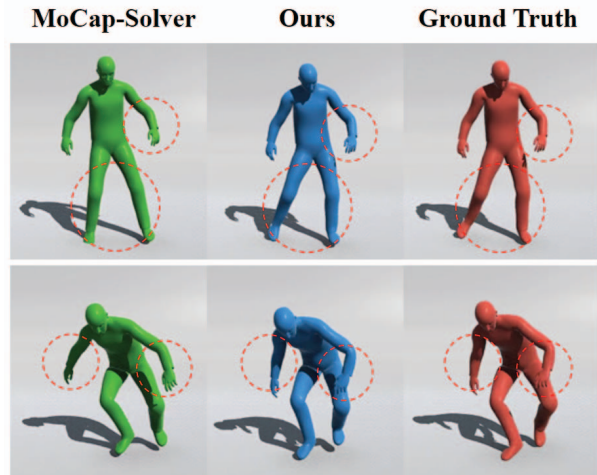


Figure 2: Visual comparisons on the real MoCap data. Compared with the state-of-the-art MoCap-Solver [1] (green), our method (blue) generates more accurate results compared to the ground-truth (red).

3 DISCUSSION

Our divide-and-conquer strategy employs local feature operators, e.g., 2D local convolution or window based multi-head self-attention (WMHSA), to learn different local sub-motions from the corresponding marker subsets within the local spatio-temporal scopes, and then aggregates them into the global motion representation to achieve accurate motion estimation. In this way, different local operators are capable of effectively representing different local motion patterns w.r.t. different subsets of markers. Compared to prior works that use the Multilayer Perceptron (MLP) or FC layers to directly predict the overall human skeleton motion from all markers, our method reduces the complexity of model learning, and generates more accurate motion results compared to the ground-truth (see Fig. 2). Besides, using local operators rather than the MLP or FC layers can significantly reduce model parameters and computational costs, making our model more lightweight and achieving better real-time performance for practical applications. As future work we would like to conduct extensive experiments to fully verify the effectiveness of our divide-and-conquer strategy, and further improve the result accuracy and model robustness to marker order and number.

4 CONCLUSION

In this paper, we propose a divide-and-conquer strategy based network to effectively estimate human skeleton motions from sequential raw markers in real-time. We decompose the task of directly learning the overall motion of human body from all markers into first learning local sub-motions from the corresponding subsets of markers with local feature operators, and then aggregating them to obtain a global motion representation. In this fashion, our obtained motion representation significantly facilitates final human motion estimation.

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